Contact

www.linkedin.com/in/yoshuanavachocron (LinkedIn)
www.yoshuanava.com (Personal)

Top Skills

Robotics
Robot Programming
Artificial Neural Networks

Languages

English (Full Professional)
Spanish (Native or Bilingual)
German (Limited Working)

Certifications

Investigation 2.0
Image and video processing: From Mars to Hollywood with a stop at the hospital
Calculus Two: Sequences and Series
Machine Learning

Honors-Awards

Engineering Dean's List during the March-July semester of 2010
Academic Excellence Recognition Award
Recipient of a scholarship to attend the Latin American Conference of Informatics (CLEI) 2013, awarded by the Venezuelan National Endowment for Science, Technology and Innovation (FONACIT).
Honorific Mention on my undergraduate thesis project
Graduated as first of my class, and ranked in its top-5 percent during all the academic years 2009-2015

Publications

Visual-LiDAR SLAM with loop closure (Master thesis)

Yoshua Nava

Giving robots advanced perception to cope with ever-changing and unstructured environments.
Zurich, Switzerland

Summary

I am a Robot Perception Engineer with 7 years of experience developing and deploying robust Perception systems for mobile robots. I specialize in Simultaneous Localization and Mapping, having served as Tech Lead and main developer of Pharos (ANYbotics' SLAM engine).

I define myself as a professional through four pillars:
- Focus on leaving a positive legacy, technical and personal, wherever I go.
- Being able to consistently generate value, in different environments, with diverse people, and faced with a broad set of problems.
- Cultivating ideas, and seeing them take a life of their own.
- Continuous learning.

My career goal is to enable robots to cope with ever-changing environments and unforeseen situations, through advanced Perception capabilities.

Experience

Flexion Robotics
AI Engineer - Computer Vision
October 2025 - November 2025 (2 months)
Zurich, Switzerland

ANYbotics
Senior Perception Engineer
September 2018 - August 2025 (7 years)
Zürich Area, Switzerland

I served as the Technical Lead in Simultaneous Localization and Mapping. My work comprised technical strategy, project leadership, algorithmic development, software architecture, and performance optimization.

FrankenPharos: Lidar-Inertial SLAM

X-ICP: Localizability-Aware LiDAR Registration for Robust Localization in Extreme Environments

Our SLAM system, stood out from the competition by implementing the capabilities of global localization, robust localization in featureless environments, support for arbitrarily large maps, and efficient spatial queries. Pharos ranked 4th on the HILTI SLAM Challenge 2022, and 7th on the HILTI SLAM Challenge 2023.

Along the way, I built experience with different 'ways of working': supervising projects that led to the delivery of customer value, as well as new technology, taking new methods to a high Technology Readiness Level.

I acted as a project partner for most developments within the Perception Team, covering geometric computer vision, robot-centric voxel mapping, calibration, data-driven testing, sensor timestamping and ’legged’ (IMU+kinematic) odometry systems.

I participated in multiple cross-functional projects within our Software Department, covering soft real-time systems, ROS2 Middleware, Python/C++ SDKs, etc.

Finally, I was nominated to Employee of the Year (2021), Pioneer of the Year (2022), and Team Member of the Year (2023) by my colleagues.

KTH Royal Institute of Technology

Research engineer

May 2017 - August 2018 (1 year 4 months)

Stockholm, Sweden

1. Developed ROS packages for the laboratory's ABB YuMi and Clearpath Ridgeback robots.
2. Developed software to facilitate the realization of experiments with our physical robots and sensors.

Autonomous Mobile Robots. ETH-Zurich courses at edX.

Community Teaching Assistant

June 2015 - November 2016 (1 year 6 months)

I served voluntarily as the Teaching Assistant for the online course ”Autonomous Mobile Robots” from ETH-Zurich. I assisted students by solving any questions they had through the discussion forum.

Funeraria San Pedro CA

Informatics assistant
February 2010 - August 2016 (6 years 7 months)
Venezuela

Enterprise computer systems installation, setup and troubleshooting.

Pleiades Robotics Inc.
Computer Scientist
April 2015 - July 2016 (1 year 4 months)
Montreal, Canada Area

Developed sensor drivers and ported software to run onboard a NVIDIA Jetson TX1 for

Spiri, the company’s autonomous quadrotor

Simón Bolívar University - Mechatronics Research and Development Group
Voluntary Research Assistant
March 2015 - February 2016 (1 year)
Caracas, Venezuela

1. Evaluated a camera setup for monocular visual SLAM.

2. Developed a vision-based navigation system that segmented ground based on geometric

cues.

Simón Bolívar University - Mechatronics Research and Development Group
Intern
August 2014 - September 2014 (2 months)
Caracas, Venezuela

Performed an analysis of the state of the art in Visual Odometry with a target deploymentonboard an Autonomous Underwater Vehicle (AUV)

Universidad Católica Andres Bello
Teaching Assistant - Calculus I for engineering
October 2011 - February 2014 (2 years 5 months)
Venezuela

Teacher assistant of Calculus I for engineering during five consecutive semesters. Worked with groups of approximately 20 students two times a week, preparing exercises, motivational activities and evaluations.

Education

Stockholms universitet

Master of Science - MS, IT Project Management · (August 2022 - August 2025)

KTH Royal Institute of Technology

Master of Science (M.Sc.), Systems, Control and Robotics · (2016 - 2018)

Universidad Catolica Andres Bello

Engineer’s Degree, Informatics · (2009 - 2015)

North Carolina State University

Computer Science Summer Practicum, Embedded Systems · (2014 - 2014)